outline expand detail build semesters native google drive excel word docs slides powerpoint forms emial messages marketplace social media platform StartAssignmentsResearch StudioData StudioRubric self-checkAI-use policyAll tools tools

Subject workbenchTools every subject
usesSTEM · MathematicsSTEM · StatisticsSTEM · BiologySTEM · ChemistrySTEM · PhysicsSTEM · Computer scienceSTEM · Data scienceSTEM · EngineeringHumanities & languages · English and writingHumanities & languages · HistoryHumanities & languages · PhilosophyHumanities & languages · LanguagesHumanities & languages · Religious studiesSocial sciences & business · PsychologySocial sciences & business · SociologySocial sciences & business · Political scienceSocial sciences & business · EconomicsSocial sciences & business · AnthropologySocial sciences & business · Business and managementSocial sciences & business · Finance and accountingSocial sciences & business · EducationSocial sciences & business · CommunicationSocial sciences & business · Public policySocial sciences & business · Legal studiesSocial sciences & business · Criminal justiceArts & media · Art and designArts & media · MusicArts & media · Theatre, film and mediaHealth & clinical · NursingHealth & clinical · Public healthHealth & clinical · Kinesiology and nutritionField & environment · Environmental and earth science
Case analysis template In the toolkit
Facts, stakeholders, framework, alternatives, recommendation
Open Case analysis template
Spreadsheet Opens in Semester
A grid out as Excel or CSV
Open Spreadsheet
Diagram builder Opens in Semester
Flows, timelines and matrices from a description
Open Diagram builder
Research Studio In the toolkit
Question, search plan, screening, evidence matrix, citation audit
Open Research Studio
Data Studio In the toolkit
Dictionary, cleaning log, describe, chart with a table alternative, bounded conclusions
Open Data Studio
Assignment workspace In the toolkit
Stages, deliverables, rubric self-check and submission checklist
Open Assignment workspace

Rubric self-check In the toolkit
A rubric criterion turned into a checklist — never a grade prediction
Open Rubric self-check
AI-use declaration In the toolkit
What you used, for what, and what you checked
Open AI-use declaration
Study guide from sources Opens in Semester
Guides built only from material you select, with quotations matched
Open Study guide from sources
Practice paper Opens in Semester
A timed paper marked against a key
Open Practice paper
Sources Opens in Semester
Your readings with what each is for, out as BibTeX
Open Sources
Write a document Opens in Semester
Memo, report or handout out as a Word file
Open Write a document
Check the writing Opens in Semester
Grammar read back, quotations checked against the reading
Open Check the writing
Make a deck Opens in Semester
A real PowerPoint file
Open Make a deck
Group work Opens in Semester
Who has which part, and whether it lands
Open Group work

Build this as Semester's **Workspace**: a native productivity, learning, communication, and campus ecosystem—not a directory of links to Google or Microsoft.

Semester should own the working objects: files, documents, spreadsheets, presentations, forms, assignments, research collections, datasets, conversations, communities, and listings. Those objects should share one account, permission system, search, Action Center, education graph, and policy-controlled AI layer.

Below is an expanded product and implementation specification. It is a proposed build plan, not a claim that these capabilities are already implemented or deployed.

1. Evidence baseline and build constraints

The project's native-platform execution specification already calls for native productivity and collaboration, student workspaces, learning tools, community, marketplace, a shared policy layer, and controlled migration from external systems. It explicitly prohibits duplicate applications, duplicate data models, static mockups replacing working behavior, and claims of completion without evidence.[1]

The architecture snapshot dated October 4, 2026 reports mixed server-backed and device-state domains, marketplace gated off and not built, platform modules not yet adopted, and unresolved tenancy and AI-policy risks. Treat those as dated findings to reverify—not as a live assessment of today's deployment. [2]

The inspected student-screen template demonstrates reusable shell components, source-labelled cards, context bars, and design tokens. Its sample content and action labels do not establish functional document editing, export, persistence, or submission behavior. [3] [4]

Build implication:

- Reuse the existing application and shared platform primitives.
- Complete existing workflows before adding parallel versions.
- Keep comprehensive v1 scope, but activate each capability through its own evidence gate.
- Track designed, implemented, tested, deployed, institution-authorized, and production-enabled separately.
- Do not activate new sharing or AI workflows on an unverified security foundation.

2. Product structure: one workspace, multiple native tools

Primary navigation

Area	Purpose	Main destinations
Start	Understand and begin the next useful action	Continue working, upcoming deadlines, recent files, assigned group work
Assignments	Complete work from brief to submission	Brief, stages, deliverables, sources, rubric, declaration, submission
Workspace	Create and organize working materials	Files, Docs, Sheets, Slides, Forms, Diagrams, Projects
Research Studio	Develop source-supported research	Question, search plan, screening, evidence matrix, citation audit
Data Studio	Analyze data reproducibly	Dataset, dictionary, cleaning, analysis, charts, conclusions
Study	Learn from selected materials	Study guides, practice papers, flashcards, review sessions
Subject workbench	Apply discipline-specific methods	Subject templates, calculators, analytical workflows
Communications	Coordinate and communicate	Messages, channels, email, announcements
Community	Participate in campus life	Organizations, posts, events, study groups, mentoring
Marketplace	Exchange approved goods and services	Listings, offers, transactions, disputes
All tools	Find a tool by task or discipline	Searchable, filterable capability catalog

These are proposed information-architecture labels, not verified application routes.

Shared object header

Every tool should display:

- Object title and type.
- Course, project, or assignment context.
- Owner and collaborators.
- Privacy and sharing scope.
- Source and authority label.
- Last saved version and sync state.
- Applicable AI-use policy.
- Export, share, history, and recovery controls.

An assignment-linked spreadsheet should remain the same spreadsheet when opened from Files, Data Studio, or the assignment—not become three disconnected copies.

3. Native productivity suite

Google Drive, Excel, Word, Docs, Slides, PowerPoint, and Forms are functional reference points. Semester's product names should remain its own.

Semester Files: native storage and organization

Build:

- Personal, course, project, organization, and institution workspaces.
- Folders, collections, tags, favorites, recent items, and pinned materials.
- Uploads, previews, bulk operations, trash, restore, and version history.
- Search across authorized content and metadata.
- Ownership transfer and collaborator management.
- Named-recipient sharing and optional expiring links.
- Storage quotas, retention rules, and institutional legal-hold handling.
- Import/export jobs with manifests and item-level results.

Important distinction:

- Folder: organizational containment.
- Collection: references to objects without duplicating them.
- Sharing grant: explicit authorization, not an implication of appearing in search.

Acceptance: upload, reopen on another device, restore a previous version, revoke a recipient's access, and verify that unauthorized search results reveal neither content nor metadata.

Semester Docs: Word and Google Docs-style work

Build:

- Structured rich-text editing.
- Headings, tables, images, equations, footnotes, citations, and references.
- Comments, suggestions, review resolution, and version comparison.
- Templates for essays, memos, reports, handouts, case analyses, and lab reports.
- Source-linked quotations and evidence annotations.
- Accessible reading order and export checks.
- DOCX, PDF, Markdown, and plain-text export.
- DOCX import with a visible unsupported-feature report.

Do not promise lossless Word compatibility. Define a supported feature matrix and test it with representative documents.

Semester Sheets: Excel-style work

Build:

- Workbooks, sheets, cells, ranges, and named ranges.
- Formula evaluation with a documented supported-function catalog.
- Sorting, filtering, validation, conditional formatting, and protected ranges.
- Charts, descriptive summaries, pivot-style analysis, and table views.
- Units, missing-value conventions, and source-linked inputs.
- CSV and XLSX import/export.
- Dataset attachment to Data Studio.
- Formula explanations that distinguish computed results from AI commentary.

Security boundary: imported formulas must not execute arbitrary code or initiate external requests. Unsupported macros and external links should remain disabled and visibly flagged.

Acceptance: test formula dependencies, circular references, dates, locale-sensitive values, large imports, and spreadsheet-export injection cases.

Semester Slides: PowerPoint-style work

Build:

- Slide canvas, themes, layouts, and slide masters.
- Text, shapes, images, tables, charts, equations, and speaker notes.
- Alignment, grouping, ordering, and presentation mode.
- Collaborative review and version history.
- Source attribution on slides and in notes.

- Alt text and reading-order review.
- Editable PPTX export and PDF handouts.

A “real PowerPoint file” must preserve the declared supported elements as editable objects—not merely export screenshots inside a presentation.

Semester Forms

Build:

- Question types, validation, sections, and branching.
- Draft, preview, publish, close, and archive states.
- Anonymous, named, and institution-authenticated response modes.
- Consent language and response-retention settings.
- Response tables and Data Studio handoff.
- Registration, feedback, peer review, intake, and approved research templates.
- Versioned forms so later question changes do not silently reinterpret earlier responses.

Anonymous must have a precise implementation definition. Do not label responses anonymous while retaining identifying data accessible to ordinary form owners.

Semester Diagrams

Build:

- Flowcharts, timelines, concept maps, process maps, matrices, and entity diagrams.
- Manual editing after generation from a description.
- Connections to assignment stages and research evidence.
- SVG, PNG, and PDF export.
- Text-based alternatives and keyboard-editable structure.

AI-generated diagrams remain editable drafts, not validated accounts of a process.

4. Expand the supplied toolkit into complete workflows

Tool	Native workflow	Required output	Boundary
Case analysis	Facts → stakeholders → framework → alternatives → recommendation	Structured case document and evidence map	Separate established facts from assumptions
Spreadsheet	Create/import → model → calculate → review → export	Workbook, XLSX, CSV	Flag unsupported formulas and features
Diagram builder	Describe → generate → edit → validate	Editable diagram plus text alternative	Generation is not process verification
Research Studio	Question → search plan → screen → extract → synthesize → audit	Research protocol, evidence matrix, bibliography	Never invent source access or citations

Tool	Native workflow	Required output	Boundary
Data Studio	Dictionary → clean → describe → analyze → chart → interpret	Analysis record, tables, charts, bounded conclusions	Expose transformations and limitations
Assignment workspace	Brief → stages → deliverables → check → submit	Versioned deliverable bundle	Drafts are not official submissions
Rubric self-check	Criterion → checklist → evidence → unresolved gaps	Criterion-level review	Never a grade prediction
AI-use declaration	Record use → purpose → edits → verification → export	Reviewable declaration	Do not imply automatic knowledge of outside AI use
Study guide from sources	Select materials → extract → organize → verify	Source-linked guide	Preserve quotation locations
Practice paper	Scope → generate/select → attempt → mark → review	Attempt record and feedback	Key-based feedback is not an official grade
Sources	Add → deduplicate → annotate → cite → export	Source library and BibTeX	Distinguish metadata from verified full text
Write a document	Select template → draft → cite → review → export	Editable document and DOCX	Apply course AI policy
Check the writing	Select checks → review suggestions → verify quotations	Review report and accepted changes	No unsupported authorship judgments
Make a deck	Outline → compose → cite → rehearse → export	Editable presentation and PPTX	Preserve attribution and accessibility
Group work	Allocate → coordinate → review → assemble → hand off	Shared plan and deliverables	Activity logs do not measure contribution quality

The central assignment workflow

Use one connected sequence:

1. Import or create the assignment brief.
2. Preserve the official source, deadline, rubric version, and submission destination.
3. Create stages and deliverables.
4. Attach sources, datasets, Docs, Sheets, Slides, and diagrams.
5. Assign group responsibilities where appropriate.
6. Apply the course and assignment AI policy.
7. Run student-controlled self-checks.
8. Assemble a deliverable manifest.
9. Export or submit through an authorized channel.
10. Display the actual result and retain the receipt.

Recommended states:

`draft → working → ready_for_review → ready_to_submit → submitting → submitted`

Add explicit `submission_failed`, `receipt_unverified`, and `resubmission_required` states. Never infer successful submission from an export or button click.

5. Subject workbench: every listed discipline

Build subject packs on shared editors, evidence objects, datasets, and review workflows. Do not create a separate application per subject.

Subject	Specialized workbench	Deliverables and checks
Mathematics	Equation notebook, proof planner, graphing, symbolic/numerical exploration	Proofs, derivations; assumptions and domain checks
Statistics	Study design, distributions, estimation, hypothesis tests, regression	Analysis report; assumptions and uncertainty
Biology	Experiment planner, pathway diagrams, observation records	Lab report; units, controls, source-supported interpretation
Chemistry	Stoichiometry, reaction tables, structure annotation	Calculations and reports; units and approved safety references
Physics	Dimensional analysis, simulations, uncertainty propagation	Models and lab reports; boundary conditions
Computer science	Code workspace, algorithm tracing, tests, complexity notes	Code and technical reports; isolated execution
Data science	Dataset profiling, notebooks, model evaluation	Reproducible pipeline; leakage and bias checks
Engineering	Requirements, design alternatives, calculations, verification	Design report; assumptions and verification evidence
English and writing	Argument mapping, close reading, drafting, revision	Essays; quotation and citation checks
History	Chronologies, primary-source analysis, historiography	Historical argument; provenance and context
Philosophy	Argument reconstruction, objections, logic	Argument analysis; validity versus truth
Languages	Vocabulary, grammar practice, aligned texts, pronunciation practice	Practice records; instructor-controlled assessment support
Religious studies	Comparative frameworks, textual analysis, context	Analysis; tradition and interpretation distinctions
Psychology	Study design, operational definitions, evidence appraisal	Research report; ethics and causal limits
Sociology	Social theory, coding, social-pattern analysis	Analysis; sampling and contextual limitations
Political science	Institutional comparison, policy analysis, electoral datasets	Comparative report; source and measurement checks

Subject	Specialized workbench	Deliverables and checks
Economics	Models, elasticity, market analysis, econometric workflows	Model/report; assumptions and causality limits
Anthropology	Field-note organization, qualitative coding, reflexive memos	Ethnographic analysis; participant protections
Business and management	Cases, strategy, operations, stakeholder analysis	Recommendation; alternatives and tradeoffs
Finance and accounting	Financial models, statements, reconciliations	Workbook/report; assumptions and balanced calculations
Education	Lesson planning, objectives, assessment alignment	Teaching plan; accessibility and learner-data privacy
Communication	Audience analysis, message strategy, discourse analysis	Campaign/report; attribution and consent
Public policy	Problem framing, options, implementation, evaluation	Policy memo; evidence quality and distributional effects
Legal studies	Issue spotting, case briefs, authority mapping	Educational analysis; jurisdiction and source date
Criminal justice	Case/policy analysis, evidence review, system mapping	Report; uncertainty and rights-sensitive interpretation
Art and design	Moodboards, critique, iterations, asset provenance	Portfolio; licensing and accessible presentation
Music	Score annotation, theory, practice planning, listening analysis	Analysis/performance reflection; rights and provenance
Theatre, film and media	Scripts, storyboards, shot lists, production planning	Production package; permissions and asset rights
Nursing	Educational care-plan templates and simulation reflection	Simulation work; no unapproved patient information
Public health	Population measures, study appraisal, program evaluation	Public-health report; denominator and privacy checks
Kinesiology and nutrition	Measurement logs, movement analysis, literature appraisal	Educational analysis; no autonomous clinical prescriptions
Environmental and earth science	Field logs, spatial datasets, sampling, systems models	Field report; coordinate, unit, and sampling checks

Subject-pack contract

Each pack should declare:

- Supported tasks and prerequisites.
- Required input structure.
- Applicable templates and tools.
- Calculation or analysis engine.

- Permitted data classifications.
- AI permissions.
- Source and citation requirements.
- Safety limitations.
- Accessible alternatives.
- Acceptance fixtures.
- Faculty review owner.

Clinical, legal, engineering-safety, and human-participant workflows require qualified review before activation in consequential settings.

6. Research Studio and Data Studio

Research Studio

Core objects:

- Research project.
- Question and protocol versions.
- Search runs and exact queries.
- Source records.
- Screening decisions and reasons.
- Extracted evidence with page or section locators.
- Claims linked to supporting or conflicting evidence.
- Citation-audit findings.

Required behavior:

- Distinguish a search result from a retrieved source.
- Distinguish abstract-only review from full-text review.
- Preserve exact quotations separately from summaries.
- Record excluded sources and reasons.
- Mark unsupported claims rather than inventing references.
- Revalidate permissions when retrieving shared sources.

Data Studio

Core objects:

- Dataset and dataset version.
- Variable dictionary.
- Import manifest.

- Cleaning steps.
- Analysis specification.
- Executed analysis run.
- Result tables and charts.
- Interpretation and limitations.

Required behavior:

- Preserve original data.
- Make every transformation inspectable.
- Record engine and dependency versions.
- Provide an underlying table for every chart.
- Distinguish descriptive association from causal conclusions.
- Identify unsupported analyses instead of substituting fabricated results.
- Run executable analysis in bounded, resource-limited environments.

7. Native email and messages

Messages first-class inside Semester

Build:

- Direct messages and group conversations.
- Course, project, organization, and support channels.
- Threads, replies, attachments, mentions, and search.
- Notification preferences, mute, block, report, and moderation.
- Context links to assignments and shared objects.
- Accessible message composition and attachment previews.

Message membership must not automatically grant access to attached files. Show permission mismatches before sending.

Semester Mail

Design two separate capabilities:

1. Connected mail: explicitly authorized access to existing email providers.
2. Native mail: Semester-managed addresses, mailboxes, sending, receiving, and delivery operations.

Native mail requires its own delivery, abuse, domain, identity, retention, and operational readiness gates. Do not treat an account alias or compose screen as a functioning email service.

Workflow:

`draft → recipient resolution → attachment access review → confirmation → queued → delivery result`

Keep message delivery, external email delivery, and official institutional notice status distinct.

8. Community and social platform

Native community functions

- Profiles with student-controlled visibility.
- Campus, course, organization, and interest communities.
- Posts, comments, reactions, and media.
- Study groups, mentoring, events, and opportunity sharing.
- Following and discoverability controls.
- Chronological feeds and transparent recommendation settings.
- Reporting, moderator review, appeals, and restrictions.

Privacy boundaries

- Joining a university must not publish course enrollment automatically.
- Grades, disability information, advising records, and private study activity stay outside social profiles.
- Portfolio publication requires an explicit action.
- Community membership does not authorize AI access to private materials.
- Moderation access must be scoped and auditable.
- Blocking behavior must be tested across feed, search, messaging, and mentions.

Use a shared Files layer for assets, but a separate publishing permission. “Stored” and “publicly posted” are different states.

9. Marketplace: separate the two business models

Campus marketplace

For permitted goods and services:

- Listings, categories, search, favorites, and seller profiles.
- Listing moderation and prohibited-item rules.
- Offers, reservations, and transaction-linked conversations.
- Payment-provider handoff where enabled.
- Cancellation, refund, dispute, and support workflows.
- Safe handoff information with location privacy.

Exclude academic cheating services, unauthorized course materials, credentials, and institutional-access resale.

Developer marketplace

For approved applications and integrations:

- Developer registration.
- Capability and data-scope manifest.
- App review and security evidence.
- Institution-controlled installation.
- Explicit grants and revocation.
- Versioning, deprecation, metering, and incident handling.
- Installation audit and uninstall/export behavior.

The existing project specification includes a governed developer marketplace with scoped tenant installs; the dated architecture snapshot says marketplace was gated off and not built. These support the planned direction, not a readiness claim. [1] [2]

10. Shared architecture and data contracts

The project architecture identifies a modular monolith, API gateway, policy decision point, audit/outbox pattern, AI gateway, sync gateway, and integration hub. Reconcile these existing components before introducing replacements. [2]

Proposed logical model

Map these concepts onto existing canonical entities before adding tables:

Entity	Responsibility
Workspace	Personal, group, course, or institutional context
Artifact	Stable identity for a document, workbook, deck, form, diagram, or file
ArtifactVersion	Immutable saved version
Membership / AccessGrant	Relationship and explicit permissions
SourceRecord	Provenance, authority, locator, freshness
AssignmentLink	Connection to a task and deliverable requirement
Review	Comments, suggestions, rubric evidence, resolved findings
Dataset / AnalysisRun	Reproducible analytical inputs and outputs
Submission	Exact delivered versions and verified result
Conversation / Message	Communication membership and content
Publication	Explicit community-visible release

Entity	Responsibility
Listing / Transaction	Marketplace state
ToolDefinition	Capability, inputs, outputs, policy, readiness
AIUseRecord	Reviewable record of in-product AI assistance

Mutation contract

For sensitive or shared mutations:

`authenticate → resolve tenant/context → authorize → validate source/version → apply command → audit → emit event → return receipt`

Require:

- Server-resolved identity and tenant context.
- Idempotency for retryable commands.
- Version checks for conflicting edits.
- Authorization on reads, search, previews, exports, and subscriptions.
- Authorized background workers.
- Cancellation and retry behavior for long-running jobs.
- Audit logging without unnecessary private content.

Personal work needs an explicit personal ownership context. Do not invent an institutional tenant merely to store student-private artifacts.

11. Tool registry and honest interface copy

“All tools” should come from one capability registry.

Example proposed contract:

```
type ToolDefinition = {
  id: string;
  name: string;
  category: string;
  subjects: string[];
  supportedInputs: string[];
  supportedOutputs: string[];
  requiredCapabilities: string[];
  policyContext: "personal" | "course" | "institution";
  aiMode: "none" | "optional" | "required";
  availability: "available" | "preview" | "restricted" | "planned";
  evidenceRefs: string[];
};
```

Use truthful labels:

- “Open in Semester” only for a functioning native workflow.
- “Export DOCX” only where the export path is verified.
- “Preview” for incomplete functionality.
- “Requires course approval” for policy-gated capabilities.
- “Planned” where no functional implementation exists.

Each card needs a real next action, prerequisites, output description, privacy behavior, and recovery state—not repeated decorative “Open” buttons.

12. AI-use policy and rubric self-check

Policy evaluation

Resolve applicable institution, course, assignment, data, and user constraints before each AI operation. Undefined or conflicting permissions should fail closed for restricted operations.

Policy must be checked at the model and tool call—not only in interface visibility. The dated architecture report specifically identified tenant AI-policy enforcement as unresolved at the call boundary. [2]

AI-use declaration

Record:

- In-product tool used.
- Purpose.
- Input scope.
- Output accepted or rejected.
- Student edits.
- Verification performed.
- Student-entered outside-tool use.
- Applicable assignment policy.

Let students review the declaration before attaching it to a submission.

Rubric self-check

Use criterion-level states:

- Evidence attached.
- Evidence incomplete.
- Needs student review.
- Unable to verify.
- Not applicable.

Link findings to artifact locations and explain their limits. Do not convert checklist completion into a predicted grade.

13. Build sequence, proposed owners, and activation gates

All workstreams remain in scope. These phases order dependencies rather than remove ambition.

Phase	Work	Proposed accountable owner	Activation evidence
0	Reconcile existing tools, schema, routes, handlers, tests, exports	Engineering lead	Evidence-backed capability inventory
1	Harden identity, ownership, policy, Files, versions, search	Platform/security leads	Isolation, revocation, restore, and authorization tests
2	Docs, Sheets, Slides, Forms, Diagrams	Workspace lead	Persistence, supported-format fixtures, accessibility
3	Assignment workspace, sources, research, data, rubric, declarations	Learning/AI leads	Policy enforcement and end-to-end task evidence
4	Subject packs and group work	Discipline product lead	Reviewed templates and validated analytical outputs
5	Messages and mail	Communications lead	Delivery, permission, abuse, retention, and recovery evidence
6	Community and campus marketplace	Trust/community lead	Moderation, privacy, dispute, and support readiness
7	Developer marketplace and external migration	Platform/institution lead	Scoped installs, reconciliation, rollback, approved cutover

First build package

The next implementation artifact should contain:

1. Existing-versus-proposed capability inventory.
2. Canonical object and ownership map.
3. End-to-end Files → Docs → Assignment workflow specification.
4. Tool registry with evidence-linked availability.
5. Supported import/export feature matrices.
6. Permission and threat model.
7. Acceptance fixtures and failure cases.
8. Migration and rollback plan.
9. Separate readiness register.

Do not claim existing persistence or exports from screen appearance alone.

14. Release acceptance checklist

A capability is not complete until it passes the relevant checks:

- A user can create, save, reopen, edit, and recover work.
- Permission revocation affects content, previews, search, exports, and collaboration.
- Unauthorized users cannot discover restricted object metadata.
- Concurrent edits preserve work or expose a recoverable conflict.
- Imports identify unsupported features instead of silently discarding them.
- Exports open in the declared target applications with verified supported structure.
- Charts have table alternatives.
- Keyboard and screen-reader workflows are tested.
- AI-disabled contexts cannot make forbidden model or tool calls.
- Rubric reviews do not imply grades.
- Submissions preserve exact versions and show the actual receipt status.
- Publication never silently exposes private learning records.
- External sends show resolved recipients and exact content before confirmation.
- Restore, incident response, support ownership, and rollback are demonstrated.
- Institutional authority changes require explicit approved cutover evidence.

15. One integrated workflow to prove the system

Use a biology lab report as an initial cross-tool acceptance scenario:

1. Open the assignment with its official brief and rubric.
2. Organize approved readings in Sources.
3. Import measurements into Sheets.
4. Validate units and missing values in Data Studio.
5. Produce a chart and accessible data table.
6. Draft the report in Docs.
7. Link quotations and claims to source locations.
8. Coordinate review in Group work and Messages.
9. Run the rubric self-check.
10. Review the AI-use declaration.
11. Export the supported DOCX deliverable.
12. Submit through an authorized workflow and verify the receipt.

That scenario should prove the shared objects, permissions, provenance, analysis, exports, and submission lifecycle. Then add equivalent fixtures for quantitative, humanities, creative, clinical-simulation, and policy work.

The goal is not fifteen isolated tools. It is one native working environment where students can move from assignment to evidence, analysis, writing, collaboration, and delivery without losing context—or control of their work.

⁂

1. The Main Semester design system (2) copy/uploads/provide prompt comand and code to excute and go al.pdf
2. The Main Semester design system (2) copy/uploads/Semester build Architecture.pdf
3. The Main Semester design system (2) copy/templates/student-screen/StudentScreen.dc.html
4. The Main Semester design system (2) copy/templates/student-screen/ds-base.js
5. The Main Semester design system (2) copy/templates/student-screen/support.js
6. projects.semester.student_faculty_os
7. projects.semester.platform_vision
8. https://www.perplexity.ai/search/7c841de3-c8fc-46fe-8694-ad08f957487b
9. https://www.perplexity.ai/search/57b43e07-422e-4033-b258-19cc98f93d85